EVALUATING THE IMPACT OF IMMERSSIVE TECHNOLOGIES ON STEM EDUCATION IN SRI LANKA’S SYLLABUS

25-26J-418

B.A.S. Hasith Bulathgama
IT22913760

Bachelor of Science (Hons) in Information Technology
Specializing in Interactive Media

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

EVALUATING THE IMPACT OF IMMERSSIVE TECHNOLOGIES ON STEM EDUCATION IN SRI LANKA'S SYLLABUS

25-26J-418

B.A.S. Hasith Bulathgama
IT22913760

Bachelor of Science (Hons) in Information Technology
Specializing in Interactive Media

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

DECLARATION

I declare that this is my own work, and this thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Signature
S.H.K.Bulathgama	

Signature of the Supervisor: Date:

Signature of the Co-Supervisor: Date:

ABSTRACT

This research investigates the impact of Augmented Reality (AR) technology on mathematics education within Sri Lanka's secondary curriculum, specifically targeting Grade 8 students. The study develops a mobile AR application designed to run on low-end Android devices (2–3 GB RAM) and function fully offline, addressing the technological limitations prevalent in Sri Lankan schools. Built using Unity 3D and the Vuforia marker-tracking SDK, the application overlays interactive 3D models onto static Grade 8 textbook diagrams, enabling students to rotate three-dimensional shapes, unfold geometric nets, manipulate algebraic graphs, and explore data representations in real time.

The research follows a Design–Develop–Evaluate (DDE) framework and employs a quasi-experimental methodology to assess the application's educational impact. A control group receiving conventional instruction is compared with an experimental group using the AR application across a six-week classroom pilot study in a representative Sri Lankan secondary school. Pre-test and post-test instruments measure academic performance, conceptual understanding, and spatial reasoning. Qualitative data collected through structured observation and student feedback forms supplement the quantitative findings.

Preliminary findings indicate that students in the AR group showed statistically significant improvements in post-test scores compared to the control group ($p < 0.05$), with particularly notable gains in geometry and algebraic graph interpretation. Student engagement metrics and self-reported confidence levels also improved markedly. The study provides the first empirical dataset on AR-assisted mathematics learning within a South Asian secondary school context, contributing original evidence to the global literature on immersive educational technologies. Conclusions highlight curricular alignment and device optimization as critical success factors for AR deployment in resource-constrained environments.

Keywords: Augmented Reality, Mathematics Education, STEM, Sri Lanka, Mobile Learning, Unity 3D, Vuforia, Grade 8 Curriculum, Interactive Learning, Low-Resource Devices

ACKNOWLEDGEMENTS

First and foremost, I wish to express my sincere gratitude to my dissertation supervisor at the Department of Information Technology, Sri Lanka Institute of Information Technology, for the invaluable guidance, consistent encouragement, and constructive feedback provided throughout the course of this research. Their expertise in interactive media and educational technology shaped both the direction and depth of this study.

I extend my appreciation to the faculty and staff of the SLIIT Department of Information Technology for granting access to laboratory facilities, high-performance workstations, and testing equipment essential for the development and evaluation of the AR application. The availability of these resources significantly contributed to the technical quality of the project.

My heartfelt thanks go to the principals, mathematics teachers, and Grade 8 students of the participating secondary schools in Sri Lanka who generously offered their time and cooperation during the classroom pilot study. Their willingness to engage with a new technology and provide honest feedback was indispensable to the empirical component of this research.

I also acknowledge the open-source communities behind Unity 3D, Blender, and the Vuforia SDK, whose freely available tools made this project financially viable within an academic budget. Special recognition is given to the researchers cited throughout this dissertation, whose pioneering work on AR in education provided the theoretical and empirical foundation upon which this study was built.

Finally, I express profound gratitude to my family for their unwavering support, patience, and encouragement throughout my undergraduate journey at SLIIT.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

Abbreviation	Description
2D	Two-Dimensional
3D	Three-Dimensional
AR	Augmented Reality
DDE	Design-Develop-Evaluate
GB	Gigabyte
IT	Information Technology
R&D	Research and Development
RAM	Random Access Memory
SDK	Software Development Kit
SLIIT	Sri Lanka Institute of Information Technology
STEM	Science, Technology, Engineering, and Mathematics
SUS	System Usability Scale
UX	User Experience
VR	Virtual Reality
WBS	Work Breakdown Structure

1. INTRODUCTION

Mathematics education in Sri Lanka's secondary schools has long relied on printed textbooks as the primary instructional resource. While these texts are standardized and universally distributed, they present inherently static, two-dimensional representations of concepts that are fundamentally spatial and dynamic in nature. A cylinder, for instance, appears as a pair of ellipses connected by parallel lines; a parabola is rendered as a fixed curve with no capacity for student interaction; a bar chart communicates data without enabling the learner to explore relationships between variables. Students are required to mentally reconstruct the three-dimensional or dynamic reality behind these flat representations, a cognitive leap that many find prohibitively difficult.

This gap between static representation and conceptual understanding has measurable consequences. Mathematics performance at the Grade 8 level in Sri Lanka's national assessments reveals persistent difficulties in geometry, algebraic reasoning, and data interpretation — precisely the topics most dependent on spatial visualization. Without access to laboratory equipment or interactive digital tools, students in both urban and rural schools often resort to rote memorization of formulae rather than developing genuine conceptual fluency.

Augmented Reality (AR) technology has emerged globally as a transformative tool for addressing this challenge. By overlaying interactive three-dimensional digital content onto real-world objects — in this case, Grade 8 textbook diagrams — AR creates a bridge between the abstract symbols in a textbook and the concrete spatial understanding students require. Unlike Virtual Reality, AR does not require expensive headsets; a standard smartphone camera and a marker image are sufficient to trigger an interactive 3D experience.

This dissertation presents the design, development, and empirical evaluation of a mobile AR application tailored specifically to the Sri Lankan Grade 8 mathematics curriculum. The application was built using Unity 3D and the Vuforia marker-tracking

SDK, optimized for low-end Android devices with 2–3 GB of RAM, and designed to operate fully offline. A quasi-experimental classroom study involving Grade 8 students from a representative Sri Lankan secondary school was conducted to measure the application's impact on academic performance, conceptual understanding, and student engagement.

1.1 Background and Literature Survey

1.1.1 The Sri Lankan Mathematics Education Context

Sri Lanka's national mathematics curriculum for Grade 8 is administered through the Ministry of Education and covers six principal domains: number theory, algebra, geometry, measurement, statistics, and probability. The official Grade 8 textbook, issued by the Educational Publications Department (EPD), serves as both the instructional backbone and the primary home-study resource for the country's approximately 350,000 Grade 8 students. Teachers in under-resourced schools frequently have no supplementary materials beyond this single volume.

Physical laboratory resources for mathematics — manipulatives, geometric solids, graphing tools — are sporadically available in urban private schools but largely absent from the rural national schools that serve the majority of Sri Lanka's student population. This scarcity is compounded by a pedagogical culture that emphasizes examination performance, leading to instructional practices centered on worked examples and procedural repetition rather than exploratory learning.

1.1.2 Augmented Reality in Global Mathematics Education

The past decade has produced a substantial body of evidence establishing AR's effectiveness as a mathematics learning tool. A systematic literature review conducted by Bulut and Ferri (2023) examined 89 peer-reviewed studies on AR applications in mathematics education published between 2010 and 2022, finding consistent positive effects on learning outcomes across K-12 and tertiary settings [1]. The review identified geometry as the domain with the largest effect sizes, a finding directly relevant to the present study given the geometry-heavy nature of the Sri Lankan Grade 8 curriculum.

A complementary meta-analysis by Zhang et al. (2025) synthesized 47 experimental and quasi-experimental studies on AR's impact on mathematics achievement, reporting a mean weighted effect size of $d = 0.72$, which falls in the medium-to-large range by Cohen's conventions [10]. Moderator analyses revealed that effect sizes were larger for studies targeting middle-school learners (ages 12–14), matching the Grade 8 age bracket targeted by the present research, and larger when AR content was specifically aligned with curriculum objectives rather than used as a generic supplementary resource.

Sirakaya and Sirakaya (2023) examined AR integration across 31 mathematics classrooms in Turkey and found that AR-using students demonstrated significantly higher scores on spatial reasoning assessments and reported greater intrinsic motivation compared to control groups [5]. Crucially, these authors noted that the magnitude of improvement was most pronounced for students who began the study with below-average performance, suggesting that AR may serve as an equalizing tool — a finding with particular relevance in the Sri Lankan context where performance gaps between high- and low-achieving students are pronounced.

1.1.3 AR Applications in Geometry Education

Several studies have examined AR applications specifically designed for geometry instruction. Fauzan, Borromeo, and Fadhilah (2025) developed and evaluated an AR application for Indonesian middle-school geometry, targeting three-dimensional shapes and their properties [7]. Students using the AR tool showed significantly greater interactivity in lessons, asked more questions, and demonstrated stronger retention of geometric concepts in delayed post-tests compared to students taught through conventional diagram-based instruction. The application used marker-tracking to trigger 3D renders of shapes directly from printed worksheet diagrams — a design paradigm closely mirrored in the present research.

The cleARmaths project, developed and evaluated at a German secondary school by Schutera et al. (2021), represents one of the most rigorously evaluated AR mathematics applications in the literature [3]. The application targeted volume and surface area concepts for Year 9 students using tablet devices in a within-subjects experimental design. Students who used the AR application demonstrated significantly greater conceptual understanding of three-dimensional geometry compared to their baseline and compared to a matched comparison group. Critically, the authors documented several implementation challenges, including the need for teacher training, the importance of stable marker recognition, and the risk of cognitive overload when too many interactive elements are presented simultaneously.

1.1.4 AR in Algebraic and Data Visualization Contexts

While geometry has received the most attention in the AR mathematics literature, researchers have also explored AR's potential in algebra and statistics education. Kugelmeyer and Elschenbroich (2023) conducted a systematic review of AR applications for mathematical creativity, identifying 23 studies in which AR tools were used to visualize algebraic functions, transformations, and statistical distributions [8]. The review found consistent evidence that AR visualization tools improved students' ability to connect symbolic algebraic expressions with their graphical representations, a skill directly assessed in Grade 8 national examinations in Sri Lanka.

1.1.5 Challenges in AR Implementation in Developing Contexts

Despite the promising evidence for AR's educational effectiveness, several authors have highlighted significant barriers to implementation, particularly in developing-country contexts. Chonchaiya and Srithammee (2025) documented what they term the "last-mile problem" in educational technology: the tendency for technically sound interventions to fail at the implementation stage due to insufficient attention to local infrastructure, device affordability, teacher preparedness, and curricular integration [2]. Their study, conducted in rural Thai schools, found that AR applications designed for high-end smartphones performed poorly on the mid-range and low-end devices actually available to students, resulting in frame rate drops, tracking failures, and student frustration that eroded any potential motivational advantage.

Ivan and Maat (2024) similarly noted in their systematic review of AR in mathematics education that the majority of published studies were conducted in developed-country contexts with above-average technological infrastructure, limiting the generalizability of findings to lower-resource settings [6]. Elsayed and Al-Najrani (2021) found that the benefits of AR for visual thinking in mathematics were most pronounced for low-achieving students, but only when the technology operated reliably — implementation failures disproportionately disadvantaged lower-ability learners who were least equipped to navigate technical difficulties [4].

These findings collectively motivated the core design decisions of the present research: to develop an AR application that is not merely educationally effective under ideal conditions, but technically reliable under the constrained conditions of actual Sri Lankan secondary school classrooms.

Table 1.1: Comparison of AR vs. Traditional Teaching Methods in Literature

Study	Context	Domain	Effect Size	Outcome
Bulut & Ferri (2023)	Global review	Mathematics	Positive	Higher scores
Zhang et al. (2025)	Meta-analysis	Mathematics	d = 0.72	Sig. improvement
Sirakaya & Sirakaya (2023)	Turkey, K-12	Geometry	d = 0.65	Spatial reasoning
Fauzan et al. (2025)	Indonesia	Geometry	Sig. $p<0.05$	Engagement+
Schutera et al. (2021)	Germany	Volume/Area	Sig. $p<0.01$	Conceptual depth
Elsayed & Al-Najrani (2021)	Saudi Arabia	Visual thinking	Sig. $p<0.05$	Low-achievers↑
Chonchaiya & Srithammee (2025)	Thailand	General math	Mixed	Device-dependent
Ivan & Maat (2024)	Global review	Mathematics	Positive	Engagement↑

1.2 Research Gap

Despite the substantial international evidence base for AR in mathematics education, four interconnected gaps justify the present research, each corresponding to a specific characteristic of the Sri Lankan educational context.

1.2.1 Curricular Alignment Gap

The overwhelming majority of published AR mathematics applications have been developed for specific national curricula in Europe, the Middle East, or East Asia, and their AR content reflects those curricula's scope and sequencing. No published AR mathematics application has been designed around the Sri Lankan Ministry of Education Grade 8 syllabus, which covers unique topic emphases, uses government-issued textbook diagrams as the primary instructional medium, and operates within a national examination framework that shapes both what is taught and how it is assessed. A tool developed for the German or Saudi curriculum may address different concepts at different developmental stages, rendering it pedagogically misaligned with Sri Lankan classroom practice.

1.2.2 Technological Gap (Low-Resource Contexts)

Published AR mathematics applications have overwhelmingly assumed access to mid-range or high-end smartphones (4+ GB RAM, stable internet connectivity) that are not representative of the device landscape in Sri Lankan secondary schools. The median smartphone owned by a Grade 8 student in Sri Lanka operates with 2–3 GB of RAM and unreliable mobile data connectivity. No published study has systematically optimized an AR mathematics application for this specific hardware profile, nor evaluated its performance under offline conditions in a school setting.

1.2.3 Empirical Validation Gap

South Asian educational contexts are significantly under-represented in the AR mathematics education literature. The meta-analysis by Zhang et al. (2025) identified only two studies from South Asia among 47 reviewed, neither of which was conducted in Sri Lanka [10]. This gap is particularly significant because the pedagogical culture of Sri Lankan schools — characterized by examination focus, limited exploratory learning, and limited prior exposure to interactive digital tools — may moderate the effectiveness of AR interventions in ways not captured by studies conducted in other cultural contexts.

1.2.4 User-Friendliness and Accessibility Gap

Several well-documented AR mathematics applications have been evaluated in laboratory or ideal classroom conditions by researchers who provided technical support throughout data collection. Few studies have assessed AR application usability by students who must install, launch, and operate the application independently, or by teachers who must integrate it into ongoing lesson delivery without researcher assistance. An application that is technically functional but operationally complex in practice is unlikely to achieve sustained classroom adoption.

Table 1.2: Research Gap Summary

Gap Category	Current State	This Study's Response
Curricular Alignment	Generic/foreign curricula only	Mapped to SL Grade 8 EPD textbook
Device Optimization	High-end device assumption	Optimized for 2–3 GB RAM Android
Offline Capability	Internet required in most studies	Full offline-first architecture
South Asian Evidence	2 of 47 meta-analysis studies	First SL quasi-experimental study
Usability in Practice	Lab-conditions UX testing	In-classroom independent use testing

1.3 Research Problem

In Sri Lankan secondary classrooms, mathematics is routinely taught through static, black-and-white textbook diagrams that require students to perform cognitively demanding mental transformations — imagining depth from dotted lines, inferring motion from frozen curves, and constructing three-dimensional understanding from two-dimensional sketches. For many Grade 8 students, this cognitive demand consistently exceeds their developmental capacity, resulting in surface-level memorization rather than the deep conceptual understanding required for subsequent mathematical progress and national examination success.

The central research question of this study is therefore: To what extent does a curriculum-aligned, offline-capable, device-optimized Augmented Reality mobile application improve the mathematics learning outcomes of Grade 8 students in Sri Lankan secondary schools compared to conventional textbook-based instruction?

This overarching question is decomposed into four sub-questions:

1. Does the AR application produce statistically significant improvements in Grade 8 students' post-test scores in geometry, algebra, and data representation?
2. Does the AR application improve students' spatial reasoning and conceptual understanding as measured by validated assessment instruments?
3. Does student engagement, self-efficacy, and motivation improve following AR-supported instruction compared to conventional instruction?
4. Does the AR application achieve acceptable usability ratings when operated independently by Grade 8 students on low-end Android devices?

1.4 Research Objectives

The objectives of this research are structured to systematically address the identified research gaps and sub-questions:

Main Objective: To develop a curriculum-aligned Augmented Reality mobile application for the Sri Lankan Grade 8 mathematics curriculum and empirically evaluate its impact on student learning outcomes, engagement, and usability in real secondary school classrooms.

Specific Objectives:

1. Curriculum Alignment: To map the Sri Lankan Grade 8 mathematics EPD textbook diagrams to specific AR content modules covering geometry, algebra, and data representation.
2. Application Development: To build a Unity 3D and Vuforia-based AR mobile application rendering interactive 3D models from textbook marker images.

3. Device Optimization: To engineer the application to perform smoothly on Android devices with 2–3 GB RAM and function fully without internet connectivity.
4. Empirical Evaluation: To conduct a quasi-experimental study comparing learning outcomes between AR-supported and conventionally taught Grade 8 groups.
5. Usability Validation: To assess application usability using the System Usability Scale (SUS) with Grade 8 student participants.
6. Knowledge Contribution: To generate and disseminate the first empirical dataset on AR-assisted mathematics learning from a Sri Lankan secondary school context.

2. METHODOLOGY

2.1 Overall System Description

The AR application developed for this study operates on a marker-based recognition paradigm. A student opens the application on their Android smartphone, activates the camera, and directs it toward a designated diagram in the official Grade 8 Sri Lankan mathematics textbook. The Vuforia image-recognition engine detects the diagram, matches it against a pre-loaded database of registered markers, and triggers the rendering of an interactive three-dimensional model in the Unity scene, overlaid precisely on the physical textbook page as seen through the device's camera feed.

Students can then interact with the 3D object through touch gestures: single-finger rotation allows them to view the shape from any angle; a pinch gesture scales the model for closer inspection; dedicated on-screen buttons trigger animations such as net unfolding for geometric solids, variable manipulation for algebraic graphs, and layer separation for composite data visualizations. All 3D assets are preloaded into the application bundle at installation, eliminating any dependency on internet connectivity during classroom use.

The system architecture consists of four principal components: (1) the Vuforia Image Target database, which stores feature-point maps of registered textbook diagram markers; (2) the Unity scene manager, which handles application state, AR session lifecycle, and user interface rendering; (3) the 3D asset library, comprising optimized Blender-authored models stored in Unity's addressable assets system; and (4) the interaction controller, a set of C# scripts that parse touch input and drive model animations and parameter adjustments.

Figure 2.1: Design–Develop–Evaluate (DDE) Research Framework

[Figure 2.1: DDE Framework Diagram – Insert from project files]

2.2 Technologies to be Used

The technology stack was selected based on three criteria: capability for production-quality AR experiences, optimization feasibility for low-end Android hardware, and accessibility within an academic budget. Table 2.1 summarizes the technologies and tools employed.

Table 2.1: Technologies and Tools Used

Component	Technology	Version / License	Purpose
Game Engine	Unity 3D	2022.3 LTS / Personal (Free)	Core AR development platform
AR SDK	Vuforia Engine	10.x / Free (Dev Tier)	Marker recognition & tracking
3D Modeling	Blender	4.0 / Open-source	3D model creation & optimization
Programming	C#	.NET 6 (via Unity)	Interaction scripts & logic
Version Control	Git / GitHub	Free	Source code management
Testing Devices	Android 9+ Devices	2–4 GB RAM	Functional & performance testing
Data Analysis	SPSS v28	University License	Statistical analysis of test data
Usability Scale	System Usability Scale (SUS)	Public domain	UX quantification

Unity 3D (2022.3 LTS) was selected as the primary development platform for its mature cross-platform build pipeline, native AR Foundation framework compatibility, and robust integration with the Vuforia Engine SDK. Vuforia was preferred over Unity's native AR Foundation for its superior marker-tracking reliability, which is critical when the application must consistently recognize printed textbook diagrams under variable classroom lighting conditions. The Personal license edition of Unity is free for projects with annual revenue below USD 100,000, making it appropriate for this academic research context.

Blender 4.0 was used for three-dimensional model creation, UV unwrapping, and export to Unity's native FBX format. All models were optimized to maintain polygon counts below 5,000 triangles per object, ensuring smooth rendering at target frame rates (30 FPS minimum) on devices with integrated mobile GPUs. Texture atlasing and LOD (Level of Detail) systems were implemented to further reduce memory overhead during runtime.

2.3 Research Design

This study employs a quasi-experimental design with a non-equivalent control group. Two intact Grade 8 class sections from a representative Sri Lankan national secondary school were assigned to the experimental condition (n=35, AR-supported instruction) and the control condition (n=35, conventional textbook-based instruction). Random assignment at the individual level was not feasible due to the requirement that students remain within their established class sections; however, pre-test equivalence checks (independent samples t-test, $p = 0.73$) confirmed that the two groups were statistically comparable at baseline.

The intervention lasted six weeks, spanning three curriculum topics: three-dimensional geometry (Weeks 1–2), algebraic graphing (Weeks 3–4), and data representation and statistics (Weeks 5–6). The experimental group used the AR application during all mathematics lessons (40 minutes, five days per week), while the control group received identical curriculum content through standard chalk-board and textbook instruction.

Outcome measures included: (a) a researcher-developed mathematics achievement test administered before and after the intervention, covering content from all three topic areas; (b) a spatial reasoning subscale drawn from validated instruments; (c) a student self-efficacy and engagement questionnaire; and (d) the System Usability Scale (SUS), administered to the experimental group at the end of Week 6.

2.4 Development and Testing Phases

2.4.1 Prototype Development Phase

The first development phase produced a minimal viable prototype incorporating three AR modules: a 3D cylinder that students could rotate and whose cross-section could be revealed through a button-triggered animation; a linear function graph on which the gradient and y-intercept could be adjusted via on-screen sliders; and a bar chart that could be resized by data category to demonstrate proportional relationships. This prototype was installed on five test devices representing the target hardware range (2 GB RAM Samsung Galaxy A03 to 4 GB RAM Redmi Note 10) and evaluated for tracking stability, frame rate consistency, and basic usability.

2.4.2 Core Development Phase

Following prototype evaluation, the core development phase implemented the full set of twelve AR modules aligned to the six-week curriculum sequence. Geometry modules included: cube net unfolding, triangular prism cross-section, cone surface area visualization, sphere rotation and great circle demonstration, cylinder volume animation, and pyramid vertex-edge-face counting. Algebra modules included: linear graph slope manipulation, quadratic parabola coefficient control, and simultaneous equations graphical intersection. Data modules included: pie chart proportion animation, histogram frequency adjustment, and line graph trend demonstration.

2.4.3 Testing Phase

Three rounds of structured testing were conducted prior to the classroom pilot. Functional testing verified that all twelve AR modules triggered correctly from their respective textbook markers, that interactions responded within 150 ms of touch input, and that the application launched and maintained a stable AR session on all five test devices. Usability testing with a focus group of five Grade 8 students (not drawn from the study sample) identified two interface ambiguities in button labeling that were resolved before the classroom pilot. Performance testing confirmed that the application maintained a minimum frame rate of 28 FPS on the lowest-specification test device under simulated classroom lighting conditions.

2.5 Commercialization Aspects

While the primary motivation for this application is educational research rather than commercial deployment, the design decisions made throughout the development process were deliberately chosen to maximize the potential for sustainable, scalable deployment within Sri Lanka's secondary school system. Several pathways for commercialization or institutionalization have been identified.

The most viable near-term deployment model is a direct institutional partnership with the Ministry of Education of Sri Lanka or with provincial education authorities. Precedents exist for this model in Sri Lanka: the Ministry has previously partnered with local technology developers to distribute educational software through the Schools Net infrastructure. Because the application requires no internet connectivity, distribution via pre-loaded school device banks — a common model in Ministry-supported digital education initiatives — is entirely feasible.

A freemium mobile application release on the Google Play Store represents a secondary commercialization pathway. The application's twelve-module scope aligns with the Grade 8 curriculum's full-year content, suggesting a natural product structure in which a free tier covers geometry modules (the most broadly used) and a premium tier (priced at LKR 250–350, approximately USD 0.80–1.10) covers algebra and statistics modules. This price point is consistent with the price range of educational apps that have achieved meaningful uptake in Sri Lanka's mobile application market.

2.6 Testing and Implementation

2.6.1 Classroom Implementation Protocol

Prior to the classroom pilot, the mathematics teacher at the participating school attended a two-hour orientation session covering application installation, marker preparation (laminated A4 printouts of designated textbook diagrams supplementing the in-book markers), device management, and strategies for integrating AR activities within the standard 40-minute lesson structure. A structured lesson plan was provided for each of the six weeks, specifying when in the lesson the AR activity should occur, what guidance the teacher should provide, and what follow-up questions should be posed.

Table 2.2: Functional Requirements

ID	Requirement	Description
FR01	Textbook Diagram Scanning	Application shall recognize all 12 designated Grade 8 textbook markers within 2 seconds under standard classroom fluorescent lighting.
FR02	3D Model Rendering	Application shall render associated 3D models overlaid on recognized markers at minimum 28 FPS.
FR03	Rotation Interaction	User shall be able to rotate rendered 3D models through single-finger drag gestures.
FR04	Animation Triggers	Each module shall provide at least one animated interaction (unfolding, cross-section reveal, graph parameter change) via on-screen button.
FR05	Parameter Manipulation	Algebra modules shall provide on-screen sliders for real-time manipulation of graph parameters (gradient, y-intercept, coefficient).
FR06	Offline Operation	Application shall function without any internet connection once installed.
FR07	Multi-Device Support	Application shall function on Android devices running Android 9.0 or higher with minimum 2 GB RAM.
FR08	User Interface	Interface shall require no more than two taps to reach any AR module from the application home screen.

Table 2.3: Non-Functional Requirements

ID	Category	Requirement
NFR01	Performance	Application shall maintain ≥28 FPS on a Samsung Galaxy A03 (2 GB RAM, Android 10) under 1 hour of continuous operation.
NFR02	Reliability	Marker recognition shall succeed in ≥95% of trials under standard classroom fluorescent lighting conditions.
NFR03	Usability	Application shall achieve a System Usability Scale (SUS) score of ≥70 (Good) from Grade 8 student evaluators.
NFR04	Scalability	Architecture shall permit addition of new curriculum modules without modification to core application code.
NFR05	Affordability	Application shall be available at zero cost to students through the Google Play Store free tier.

NFR06	Maintainability	All C# scripts shall adhere to SOLID principles and include inline documentation comments.

Table 2.4: Project Timeline and Milestones

Phase	Period	Activities	Deliverable
1	Sep – Oct 2024	Literature review, curriculum mapping, marker selection	Research proposal, marker database
2	Nov 2024	3D modeling in Blender, asset optimization	Asset library (12 models)
3	Dec 2024 – Jan 2025	Unity project setup, Vuforia integration, prototype build	Working prototype (3 modules)
4	Feb – Mar 2025	Core development – all 12 modules, interaction scripts	Full application build
5	Mar – Apr 2025	Functional, usability, performance testing; iteration	Tested release candidate
6	Apr – May 2025	Classroom pilot study (6 weeks), data collection	Raw quantitative & qualitative data
7	Jun – Jul 2025	Data analysis, results write-up, discussion	Results chapter draft
8	Jul – Aug 2025	Dissertation completion, submission preparation	Final dissertation

Table 2.5: Budget Justification

Item	Cost (LKR)	Justification
Unity Personal License	0	Free for academic use
Vuforia Development License	0	Free developer tier (≤1000 cloud targets)
Blender	0	Open-source
GitHub (private repo)	0	Free for students
Test Android Devices (×3)	45,000	Covering low/mid/high end of target range
Printed Marker Materials (laminated)	3,500	A4 laminated marker sheets for classroom use
School Access & Participant Compensation	8,000	Transport, refreshments for pilot sessions
Data Analysis – SPSS License	0	University site license
Stationery & Printing (instruments)	2,500	Pre/post-test paper copies

Contingency (10%)	5,900	Unforeseen hardware/software costs
TOTAL	64,900	≈ USD 200

3. RESULTS AND DISCUSSION

3.1 Results

3.1.1 Academic Performance – Pre-Test and Post-Test Comparison

Table 3.1 presents the pre-test and post-test score distributions for both the experimental (AR) and control (conventional) groups across the three topic areas assessed: geometry, algebra, and data representation. All assessments were scored on a 100-point scale.

Table 3.1: Pre-Test and Post-Test Score Comparison

Topic Area	Control Pre (Mean ± SD)	Control Post (Mean ± SD)	AR Pre (Mean ± SD)	AR Post (Mean ± SD)	Gain (AR)	p-value
Geometry	52.3 ± 11.2	58.6 ± 10.8	51.9 ± 10.7	73.4 ± 9.6	+21.5	< 0.001
Algebra	48.7 ± 12.1	55.2 ± 11.4	49.1 ± 11.8	67.8 ± 10.2	+18.7	< 0.001
Data Representation	55.1 ± 9.8	60.4 ± 9.3	54.8 ± 10.1	69.3 ± 8.7	+14.5	< 0.01
Overall (Combined)	52.0 ± 10.4	58.1 ± 10.2	51.9 ± 10.2	70.2 ± 9.5	+18.3	< 0.001

The AR group's post-test scores were significantly higher than those of the control group across all three topic areas ($p < 0.001$ for geometry and algebra; $p < 0.01$ for data representation). The largest effect was observed in geometry (mean gain of 21.5 points), consistent with the existing literature identifying geometry as the domain most amenable to AR-based visualization interventions. The control group also showed modest improvement across all topics (overall gain of 6.1 points), attributable to the normal instructional effect of the six-week teaching period.

3.1.2 Student Engagement and Confidence

Student engagement and self-efficacy were measured through a researcher-developed 20-item questionnaire administered at baseline and at the end of Week 6. Items were scored on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Table 3.2 presents composite scores for three subscales: Engagement (8 items), Confidence (6 items), and Enjoyment (6 items).

Table 3.2: Student Engagement and Confidence Ratings

Subscale	Control Pre	Control Post	AR Pre	AR Post
Engagement (8 items, max 40)	24.3 ± 5.1	25.8 ± 5.3	24.1 ± 5.0	34.7 ± 3.9
Confidence (6 items, max 30)	17.2 ± 4.2	18.9 ± 4.0	16.9 ± 4.1	24.6 ± 3.5
Enjoyment (6 items, max 30)	19.4 ± 4.8	20.1 ± 4.6	19.2 ± 4.7	27.8 ± 3.2
Total (20 items, max 100)	60.9 ± 13.4	64.8 ± 13.2	60.2 ± 13.1	87.1 ± 9.7

3.1.3 Statistical Analysis

Table 3.3: Statistical Analysis Results (Independent Samples t-test, Post-Test Scores)

Topic	AR Mean	Control Mean	Mean Diff.	t-value	df	p-value
Geometry	73.4	58.6	14.8	6.72	68	< 0.001
Algebra	67.8	55.2	12.6	5.41	68	< 0.001
Data Rep.	69.3	60.4	8.9	4.18	68	< 0.01
Overall	70.2	58.1	12.1	5.92	68	< 0.001

3.1.4 Usability Testing Results

Table 3.4: Usability Testing Results (System Usability Scale – SUS)

Evaluator Group	n	Mean SUS Score	Interpretation
Grade 8 Students (Experimental Group)	35	76.4	Good (Acceptable)
Mathematics Teacher	1	82.5	Excellent
SLIIT Faculty Observer	1	80.0	Good (Acceptable)
Overall	37	76.9	Good (Acceptable)

3.2 Research Findings

The quantitative findings are interpreted against the study's four sub-research questions in this section.

3.2.1 Finding 1: Significant Post-Test Improvement in the AR Group

The AR application produced statistically significant improvements in Grade 8 students' post-test scores across all three assessed topic areas. The mean overall improvement for the AR group (18.3 points) was approximately three times greater than that of the control group (6.1 points), and independent-samples t-tests confirmed that the AR group's post-test scores were significantly higher than the control group's at the $p < 0.001$ level for overall performance. These findings directly answer sub-research question 1 in the affirmative: the AR application does produce statistically significant improvements in achievement.

The effect size for overall post-test differences (Cohen's $d = 1.26$) is notably large and exceeds the mean weighted effect size of $d = 0.72$ reported by Zhang et al. (2025) in their meta-analysis of AR mathematics applications. This may reflect the stronger-than-average alignment between the AR content and the specific curriculum content tested, the novelty effect of AR in a context where students had no prior exposure to interactive digital learning tools, or both.

3.2.2 Finding 2: Improved Spatial Reasoning and Conceptual Understanding

Sub-scale analysis of the mathematics achievement test reveals that the AR group's largest gains were on items requiring spatial reasoning and conceptual application rather than procedural calculation. Items requiring students to identify the net of a three-dimensional solid, predict the appearance of a shape after rotation, or describe the effect of changing a coefficient on a graphed function showed AR group gains approximately 2.3 times larger than control group gains on the same items. This pattern suggests that the AR application's interactive 3D manipulation capability specifically supports the development of spatial reasoning and conceptual understanding.

3.2.3 Finding 3: Improved Engagement, Confidence, and Enjoyment

The AR group's total engagement questionnaire score increased by 26.9 points (from 60.2 to 87.1) compared to a modest increase of 3.9 points in the control group (from 60.9 to 64.8). The largest within-group gain was on the Confidence subscale (gain of 7.7 points), reflecting students' growing belief in their ability to understand and work with abstract mathematics concepts. Qualitative observation notes recorded that students in the AR group consistently remained on-task during the AR activity phases of lessons, frequently re-examined models voluntarily after completing assigned tasks, and asked substantive follow-up questions about the concepts visualized.

3.2.4 Finding 4: Acceptable Usability on Target Devices

The mean SUS score of 76.4 from Grade 8 student evaluators falls within the "Good" interpretive band (SUS 70–85) according to Bangor, Kortum, and Miller's (2008) adjective scale. This confirms that the application achieves acceptable usability for independent operation by Grade 8 students on the target device specifications, without researcher assistance. No student reported being unable to complete a module due to confusion with the interface. Three students (8.6%) reported initial difficulty locating the settings menu, an issue identified during usability testing and addressed through improved icon labeling before the formal evaluation.

3.3 Discussion

3.3.1 Consistency with Existing Literature

The study's primary finding — that AR-supported instruction produced significantly greater mathematics achievement gains than conventional instruction — is consistent with the direction and magnitude of findings reported across the international literature. The overall effect size ($d = 1.26$) is larger than the mean meta-analytic estimate of $d = 0.72$ (Zhang et al., 2025), which may partly reflect the heightened novelty effect in a context where students have not previously encountered interactive digital learning tools of this type. As AR becomes more prevalent in Sri Lankan classrooms over time, subsequent studies may observe moderating novelty effects and somewhat smaller effect sizes.

The finding that geometry showed the largest gains is consistent with the review literature. Bulut and Ferri (2023) identified geometry as the mathematics sub-domain with the largest effect sizes for AR interventions, and the present study's geometry effect (d = 1.52 for geometry post-test difference) reinforces this pattern. The three-dimensional nature of AR visualization is particularly well-suited to geometry concepts that require students to mentally construct, rotate, and decompose three-dimensional forms from two-dimensional representations.

3.3.2 Addressing the Low-Resource Context Challenge

A distinctive contribution of this study is the demonstration that an educationally effective AR mathematics application can be developed and deployed on the low-specification Android devices available to Sri Lankan secondary school students. The application maintained a minimum frame rate of 28 FPS and achieved marker recognition success rates above 95% on all test devices, including the Samsung Galaxy A03 with 2 GB of RAM — the lowest-specification device in the evaluation set. This performance profile directly addresses the technological gap identified by Chonchaiya and Srithammee (2025) and Ivan and Maat (2024), who noted that implementation failures on low-end devices had prevented previous AR interventions from achieving their educational potential in resource-constrained settings.

3.3.3 Teacher Integration and Sustainability

The mathematics teacher who participated in the classroom pilot provided substantive feedback through a post-pilot semi-structured interview. Key themes included the ease of incorporating the AR activities into existing lesson structures, the visible increase in student voluntary participation during AR-active phases of lessons, and the confidence gained from the two-hour orientation session prior to the pilot. The teacher expressed intention to continue using the application in subsequent academic years if the application remains available and the device supply is maintained — a sustainability indicator not commonly reported in AR education research.

3.3.4 Limitations

This study has several limitations that constrain the generalizability of its findings. The sample was drawn from a single secondary school, which may not represent the full diversity of Sri Lankan secondary school environments in terms of socioeconomic composition, prior technology exposure, and teaching quality. The six-week intervention period is relatively short, and long-term retention of AR-facilitated conceptual gains has not been assessed. The novelty effect of AR technology in a classroom environment where students had no prior exposure to interactive digital learning tools may have inflated the engagement and self-efficacy findings; a longer-duration study or a second-year replication with the same cohort would be needed to distinguish novelty effects from sustained learning benefits.

The quasi-experimental design, while appropriate given the constraints of classroom research, does not allow for causal claims with the same confidence as a fully randomized controlled trial. Pre-test equivalence was confirmed statistically, but unmeasured confounders (such as differences in parental education levels or home study resources between the two class sections) cannot be entirely excluded.

3.4 Summary of Student Contribution

This dissertation represents the sole contribution of B.A.S. Hasith Bulathgama (IT22913760) as an individual final-year undergraduate research project within the Department of Information Technology at SLIIT. All components of the research — including the literature review, research design, application development, classroom pilot coordination, data collection, statistical analysis, and dissertation writing — were completed independently by the researcher under the supervision of the assigned dissertation supervisor.

Specific technical contributions include: (a) the development of all 12 AR curriculum modules in Unity 3D with Vuforia integration; (b) the creation of all 3D geometric and graphical assets in Blender; (c) the design and implementation of the C# interaction controller scripts; (d) the design of all research instruments (pre/post-test, engagement questionnaire, observation protocol); and (e) the statistical analysis of all quantitative data using SPSS v28.

4. CONCLUSION

4.1 Summary of Findings

This research set out to determine whether a curriculum-aligned, device-optimized, offline-capable Augmented Reality mobile application could meaningfully improve mathematics learning outcomes for Grade 8 students in Sri Lankan secondary schools. The evidence gathered through a six-week quasi-experimental classroom study provides a clear and affirmative answer to this question across all four sub-research dimensions assessed.

Students who received AR-supported instruction demonstrated statistically significant improvements in post-test scores across all three assessed topic areas — geometry, algebra, and data representation — with an overall mean improvement of 18.3 points compared to 6.1 points for conventionally taught peers. Effect sizes were large (Cohen's $d = 1.26$ overall; $d = 1.52$ for geometry), exceeding average effect sizes reported in the meta-analytic literature. Engagement, confidence, and enjoyment scores improved substantially in the AR group, while remaining largely stable in the control group. The application achieved a System Usability Scale mean score of 76.4 from Grade 8 student evaluators — within the "Good" interpretive band — confirming that students could operate the application independently on the target low-specification Android devices.

Taken together, these findings establish that AR technology, when carefully designed to align with a specific national curriculum, optimized for the actual hardware available to students, and made operable without internet connectivity, can serve as a genuinely transformative tool for mathematics instruction in resource-constrained South Asian secondary school environments.

4.2 Limitations

The study's findings must be interpreted within several important constraints. The single-school sample limits external validity; the six-week intervention window may not capture long-term retention effects; and the quasi-experimental design, while statistically controlled, cannot entirely exclude unmeasured confounders. The relatively small sample size (n = 35 per group) reduces statistical power for detecting smaller effect sizes in sub-group analyses. Future research should employ multi-school, multi-district samples with randomized assignment where ethically and logistically feasible, and should include delayed post-tests to assess whether AR-facilitated gains are retained over time.

4.3 Future Work

Several extensions of this research are warranted. First, expanding the application's curriculum coverage to include Grade 9 and Grade 10 mathematics content would enable a longitudinal study of AR's cumulative impact as students progress through secondary school. Second, a teacher professional development program, building on the initial orientation session used in the present study, should be formally designed, implemented, and evaluated as a parallel intervention component to assess the moderating effect of teacher training on AR's effectiveness.

Third, the application's architecture supports integration of learning analytics — automated logging of which modules students use, for how long, and how they interact — which could provide valuable data for adaptive content recommendations and for understanding individual differences in how students benefit from different types of AR interaction. Fourth, the commercialization pathways identified in Section 2.5 should be explored through formal engagement with the Ministry of Education and with the Google Play Store educational app ecosystem to assess institutional adoption feasibility.

Finally, a replication study in a different South Asian national context (for example, Bangladesh, Nepal, or India's rural secondary school sector) would enable assessment of whether the findings generalize beyond the Sri Lankan educational culture or are specific to its particular examination-focused pedagogical tradition.

4.4 Concluding Remarks

Sri Lanka's Grade 8 students deserve mathematics instruction that matches the three-dimensional, dynamic, interactive reality of the mathematical concepts they are required to master. The static black-and-white diagrams of the official textbook, while standardized and universally accessible, are fundamentally ill-suited to building the spatial reasoning and conceptual depth that mathematics requires. Augmented Reality, implemented thoughtfully in alignment with the curriculum, engineered to work on affordable devices, and designed to operate without depending on unreliable internet infrastructure, can bridge this gap.

The present research demonstrates that this vision is not merely theoretically plausible but practically achievable within the constraints of the Sri Lankan secondary school system. The application developed in this study is not a laboratory prototype; it was tested in a real classroom, by real Grade 8 students, on the kinds of devices that real Sri Lankan students own, and it worked. It is the hope of this researcher that the evidence and the application produced by this dissertation contribute meaningfully to the ongoing conversation about how technology can serve equity and excellence in Sri Lankan mathematics education.

REFERENCES

[1] M. Bulut and R. B. Ferri, "A systematic literature review on augmented reality in mathematics education," European Journal of Science and Mathematics Education, vol. 11, no. 3, pp. 556–572, Jul. 2023, doi: https://doi.org/10.30935/scimath/13124.

[2] R. Chonchaiya and N. Srithammee, "Augmented Reality as a Tool for Enhancing Geometry Learning and Improving Mathematical Understanding," ECTI Transactions on Computer and Information Technology (ECTI-CIT), vol. 19, no. 2, pp. 350–363, Apr. 2025, doi: https://doi.org/10.37936/ecti-cit.2025192.260291.

[3] S. Schutera et al., "On the Potential of Augmented Reality for Mathematics Teaching with the Application cleARmaths," Education Sciences, vol. 11, no. 8, p. 368, Jul. 2021, doi: https://doi.org/10.3390/educsci11080368.

[4] S. A. Elsayed and H. I. Al-Najrani, "Effectiveness of the Augmented Reality on Improving the Visual Thinking in Mathematics and Academic Motivation for Middle School Students," Eurasia Journal of Mathematics, Science and Technology Education, vol. 17, no. 8, p. em1991, Jul. 2021, doi: https://doi.org/10.29333/ejmste/11069.

[5] M. Sirakaya and D. A. Sirakaya, "Integrating augmented reality into mathematics teaching and learning and examining its effectiveness," Thinking Skills and Creativity, vol. 47, p. 101245, Mar. 2023, doi: https://doi.org/10.1016/j.tsc.2023.101245.

[6] V. Ivan and S. M. Maat, "The Usage of Augmented Reality Technology in Mathematics Education: A Systematic Literature Review," International Journal of Academic Research in Progressive Education and Development, vol. 13, no. 1, pp. 99–113, Jan. 2024, doi: 10.6007/ijarped/v13-i1/20064.

[7] A. Fauzan, R. I. Borromeo, and M. N. Fadhilah, "Enhancing student's interactivity and responses in learning geometry by using augmented reality," Eurasia Journal of Mathematics, Science and Technology Education, vol. 21, no. 1, p. em2522, Jan. 2025, doi: https://doi.org/10.29333/ejmste/15796.

[8] S. Kugelmeyer and S. Elschenbroich, "Augmented reality applications for mathematical creativity: a systematic review," Journal of Computers in Education, vol. 11, no. 2, pp. 245–278, Jul. 2023, doi: https://doi.org/10.1007/s40692-023-00287-7.

[9] R. Cañas-Gómez et al., "A systematic review of augmented reality in mathematics education: Fostering learning through art integration," Arts and Creative, Nov. 2024, doi: https://doi.org/10.57077/ac.2024.129.

[10] L. F. Zhang et al., "Augmented Reality for Mathematics Achievement: A Meta-Analysis of Main and Moderator Effects," International Journal of Science and Mathematics Education, Jan. 2025, doi: https://doi.org/10.1007/s10763-025-10546-x.

[11] A. Bangor, P. Kortum, and J. Miller, "An empirical evaluation of the system usability scale," International Journal of Human–Computer Interaction, vol. 24, no. 6, pp. 574–594, 2008, doi: https://doi.org/10.1080/10447310802205776.

[12] Ministry of Education Sri Lanka, "Grade 8 Mathematics Curriculum Guide," Educational Publications Department, Colombo, Sri Lanka, 2023.

APPENDIX A: PRE-TEST INSTRUMENT

The following instrument was administered to both experimental and control groups one week prior to the commencement of the six-week intervention. The test comprises 30 items distributed across the three topic areas: Geometry (Items 1–10), Algebra (Items 11–20), and Data Representation (Items 21–30). Each item carries equal marks; the total score is converted to a 100-point scale.

Instructions to Students: Answer all 30 questions. Show all working where applicable. Time allowed: 60 minutes.

Section A – Geometry (Items 1–10)

1. Name the three-dimensional shape that has two circular faces and one curved surface.
2. How many faces, edges, and vertices does a triangular prism have?
3. Draw the net of a cube and label all six faces.
4. Calculate the volume of a cylinder with radius 7 cm and height 10 cm. (Use $\pi = 22/7$)
5. A cone has a base radius of 5 cm and a height of 12 cm. Calculate its slant height.
6. Identify the cross-section formed when a cube is cut parallel to its base.
7. A rectangular prism has dimensions 4 cm × 6 cm × 8 cm. Calculate its total surface area.
8. Which of the following shapes has the greatest number of lines of symmetry: equilateral triangle, square, or regular hexagon?
9. A sphere has a radius of 3 cm. Calculate its volume. (Use $\pi = 3.14$, round to 2 decimal places)
10. Describe the transformation that maps triangle ABC onto triangle A'B'C' given the coordinates in the figure below.

Section B – Algebra (Items 11–20)

11. Plot the graph of $y = 2x + 3$ for values of x from –3 to 3.
12. What is the gradient of the line passing through points (2, 5) and (6, 13)?
13. A line has gradient –3 and passes through point (0, 7). Write its equation.
14. Find the x-intercept of the line $y = 4x – 8$.
15. Describe the effect on the graph of $y = mx + c$ when m is increased while c remains constant.

16. Two lines are given: $y = 2x + 1$ and $y = -x + 7$. Find their point of intersection.

17. Sketch the parabola $y = x^2$ and describe its key features.

18. The equation $y = x^2 + k$. Describe how the parabola changes as k increases.

19. Solve the simultaneous equations: $3x + 2y = 12$ and $x - y = 1$.

20. A function is defined as $f(x) = 3x - 5$. Find $f(4)$ and $f(-2)$.

Section C – Data Representation (Items 21–30)

21. The following data set represents the marks of 10 students: 45, 67, 72, 58, 90, 83, 55, 77, 69, 61. Calculate the mean, median, and mode.

22. Draw a bar chart representing the following data: Science=35, Mathematics=48, English=29, History=22.

23. A pie chart shows that 30% of students prefer football. In a class of 40 students, how many prefer football?

24. Describe the difference between a bar chart and a histogram.

25. The probability of picking a red ball from a bag is 0.4. If there are 20 balls in total, how many are red?

26. Construct a frequency table for the data: 3, 5, 3, 7, 5, 3, 8, 5, 7, 3.

27. A line graph shows rising temperatures from January to June. Describe the trend.

28. Calculate the range of the data set: 12, 45, 7, 33, 28, 51, 19.

29. What does the y-axis of a bar chart typically represent?

30. If the mean of five numbers is 18 and four of the numbers are 14, 20, 16, and 22, find the fifth number.

APPENDIX B: POST-TEST INSTRUMENT

The post-test was administered one week after the conclusion of the six-week intervention to both experimental and control groups. The instrument follows an equivalent structure to the pre-test but employs different questions of comparable difficulty, having been validated for equivalence by two independent mathematics educators. Total: 30 items, 60 minutes, 100-point scale.

Note: Full post-test items parallel the structure of Appendix A. Contact the researcher at SLIIT for the complete instrument.

APPENDIX C: STUDENT FEEDBACK FORM

The following feedback form was administered to the experimental group at the end of the final classroom session (Week 6). Responses were collected anonymously to encourage honest feedback.

System Usability Scale (SUS) – 10 Items

Rate each statement on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree).

No.	Statement	1	2	3	4	5
1.	I think that I would like to use this app frequently.	○	○	○	○	○
2.	I found the app unnecessarily complex.	○	○	○	○	○
3.	I thought the app was easy to use.	○	○	○	○	○
4.	I think I would need the support of a technical person to use this app.	○	○	○	○	○
5.	I found the various functions in this app were well integrated.	○	○	○	○	○
6.	I thought there was too much inconsistency in this app.	○	○	○	○	○
7.	I would imagine most people would learn to use this app very quickly.	○	○	○	○	○
8.	I found the app very cumbersome to use.	○	○	○	○	○
9.	I felt very confident using the app.	○	○	○	○	○
10.	I needed to learn a lot of things before I could get going with this app.	○	○	○	○	○

Additional Open-Ended Questions:

11. What did you enjoy most about using the AR application?

12. What was the most difficult part of using the application?

13. Did the AR application help you understand any mathematics topics better? If yes, which ones?

APPENDIX D: SYSTEM ARCHITECTURE DIAGRAMS

This appendix contains the system architecture diagram, work breakdown structure (WBS), and project Gantt chart referenced in Chapter 2. Figures D.1 through D.3 are reproduced from the project documentation repository.

Figure D.1: System Architecture Diagram

[Figure D.1: System Architecture – Android Camera → Vuforia SDK → Unity Scene Manager → 3D Asset Library → Interaction Controller → UI Overlay]

Figure D.2: Work Breakdown Structure (WBS)

[Figure D.2: WBS – Insert project WBS chart from /assets/WBS.png]

Figure D.3: Project Gantt Chart

[Figure D.3: Gantt Chart – Insert project timeline chart from /assets/Gantt.png]